

Resonant charging

The commonly used inductive charging system works at a lower frequency in the kHz range while resonant wireless charging works at a higher frequency range. Resonant systems do work in different frequencies from as low as 20kHz all the way to 20MHz or more. The most common standard these days for resonant charging is 6.78MHz, which has been adopted as a standard by the AirFuel Resonance.

The higher frequency allows higher power transfer, greater range, and better efficiency. It also allows charging through thick surfaces with greater spatial freedom. Unlike the lower frequency inductive charging, which has to be tightly coupled, the resonant charging does not need to have similar-sized coils and thus can operate over a wide range of coupling as it has a high-quality factor, which also means less heating of the coil.

The coil size in resonant charging can be made smaller than that of inductive charging solutions. Furthermore, the coil need not be made from any special materials and can be created with regular copper wire or built on PCB materials. The coil can also be of a different shape, which is not possible in inductive charging. The metal heating is also negligible at a higher frequency.

The main advantage of resonant charging are:

- Lower cost
- Greater spatial freedom
- Faster charging speed
- Compact size
- Lower heat loss

2015. This standard improves control of the power transfer between the transmitter and receiver and mandates hardware based authentication for higher security. The Qi 1.3 allows for higher data transfer rates up to 15W. Modules such as Renesas's P9261 and STM's STWLC98 feature Qi 1.3 standard.

Integration of antenna board

The compact transmitter and receiver modules that utilise resonant charging working at a frequency of 13.56MHz are capable of transmitting both power and data. These modules come with an integrated

TABLE 1 INDUCTIVE CHARGING VS RESONANT CHARGING		
Specifications	Inductive Charging	Resonant Charging
Frequency range	100-300kHz	Up to 20MHz
Charging distance	Up to a few cm	Up to a few inches
Multi-device charging	No	Yes
Spatial freedom	Very low	High
Power range	A few watts	20W to 20kW
Efficiency	Good (up to 80%)	High (up to 95%)
Coil size	Large	Small
Rx-Tx communication	In-band communication	BLE/In-band communication



Fig. 2: The compatibility of Qi1.3 standard (Source: Link)

antenna in a small form factor, thus reducing the size of the product along with the time and cost of development.

These boards use the same frequency band as the NFC communication standard. These modules mostly support both bi-directional data communication (256B Max. at 212kbps) and NFC Forum Type 3 Tag transmission. The bi-directional data transfer allows for advanced functions, such as firmware download, secure data transfer, and rewriting of sensor, device, or authentication information to battery output voltage values.

Modules such as ROHM BP3621, ROHM BP3622, and ML7660 from Lapis Technologies have integrated antenna boards.

More powerful processors

The wireless charging technology has been adopting new features to improve the stability, efficiency, and security of the system, thus requiring more powerful processors. The microprocessor also allows for easier tuning for certifications along

with specification updates without PCBA change. Having a powerful processor helps in making functions extensible and enhances the capability to better customise the user interface (UI). The electromagnetic compatibility (EMC) performance depends on the microprocessor and its software operations.

Renesas's P9261-3C-CRBv2 design includes a Renesas RH850 automotive microprocessor as a host controller, which enables the system to provide functional safety features that comply with automotive safety integrity level B (ASIL B). NXP MWCT1x23 is built upon the Arm Cortex-M4 core operating at up to 168MHz with DSP and a floating-point unit. STM STSAFE-V110 runs on an Arm SecurCore SC300TM 32-bit RISC core, which features a secured operating system that enhances the cryptographic security of the module. STM STWBC2 HP is a digital controller for wireless transmitters and runs on a 32-bit Arm Cortex M0 processor running at frequencies up to 64MHz.